THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The nth term of the pattern $4, 11, 18, 25, \ldots$ is $a + (n-1)d$. The 12th term is: 1

(a) 81 **(b)** 82 **(c)** 74 **(d)** 88

2. In the pattern $3, 8, 13, 18, \ldots$ which term number equals 88? 1

(a) 17 **(b)** 19 **(c)** 18 **(d)** 20

3. What is the sum of the first 12 even numbers $2 + 4 + 6 + \ldots$? 1

(a) 156 **(b)** 168 **(c)** 157 **(d)** 144

4. Which triangular number comes just after 190? 1

(a) 211 **(b)** 229 **(c)** 210 **(d)** 209

5. A pattern of stacked sticks uses 3 sticks for 1 triangle and 2 more sticks for each extra triangle added in a row. Sticks for 10 triangles: 1

(a) 21 **(b)** 23 **(c)** 22 **(d)** 20

6. The sum of the first 24 natural numbers added to the sum of the first 25 natural numbers is: 1

(a) 624 **(b)** 626 **(c)** 649 **(d)** 625

7. Powers of 2: $2, 4, 8, 16, \ldots$ The 7th term is: 1

(a) 130 **(b)** 192 **(c)** 126 **(d)** 128

8. The nth term of the pattern $9, 15, 21, 27, \ldots$ is $a + (n-1)d$. The 15th term is: 1

(a) 94 **(b)** 87 **(c)** 93 **(d)** 99

9. In the pattern $7, 11, 15, 19, \ldots$ which term number equals 103? 1

(a) 24 **(b)** 25 **(c)** 26 **(d)** 27

10. What is the sum of the first 15 even numbers $2 + 4 + 6 + \ldots$? 1

(a) 241 **(b)** 255 **(c)** 240 **(d)** 225

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the nth term formula of $5, 8, 11, 14, \ldots$ and the 20th term. 2

12. Is 36 a term of $3, 7, 11, 15, \ldots$? Give a reason. 2

13. Find the sum of the first 30 odd numbers and the first 30 even numbers. 2

14. Find the nth term formula of $2, 9, 16, 23, \ldots$ and the 15th term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How many triangles' worth of sticks: a row of 12 triangles needs 3 sticks for the first triangle and 2 more for each new one. Find the sticks needed and the number of triangles that can be made with 35 sticks. 3

16. Find the sum of all odd numbers from 1 to 59. Hence find the sum of all even numbers from 2 to 60. 3

17. Write the 20th and 21th triangular numbers and check that they add to a square. 3

OR

How many triangles' worth of sticks: a row of 15 triangles needs 3 sticks for the first triangle and 2 more for each new one. Find the sticks needed and the number of triangles that can be made with 41 sticks. 3

18. Find the sum of all odd numbers from 1 to 79. Hence find the sum of all even numbers from 2 to 80. 3

OR

Write the 24th and 25th triangular numbers and check that they add to a square. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A pattern of squares: the first figure has 5 squares and each next figure has 4 more. Find the number of squares in the 10th figure, and the first figure that has more than 91 squares. 5

OR

Show that the sum of the cubes of the first 9 numbers is a perfect square, and find the sum of the first 9 numbers. 5

20. A pattern of squares: the first figure has 3 squares and each next figure has 5 more. Find the number of squares in the 12th figure, and the first figure that has more than 108 squares. 5

OR

Show that the sum of the cubes of the first 10 numbers is a perfect square, and find the sum of the first 10 numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

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